3 Speed

The following definitions describe speed.

Speed is a measure of how fast an object is moving.

Speed is a measure of the rate at which an object moves.

Speed is a measure of the rate of change of position of an object.

Speed is measured in units, e.g. metres per second ($m\ s^{-1}$), or kilometres per hour ($km\ h^{-1}$), or centimetres per 100 years.

Imagine a car travels a distance of 80 km in 2 hours. Assuming the car travelled at the same speed (i.e. no traffic lights, hills, corners, or any other things which might cause it to slow down or speed up, or in other terms, **no acceleration**) in 1 hour the car would have travelled 40 km. Its **average speed** would be 40 $km\ h^{-1}$. It moves at an **average rate** of 40 $km\ h^{-1}$. Its **average rate of change of position** is 40 $km\ h^{-1}$. We can use the following equation to calculate the average speed of an object.

$$\text{Average speed} = \frac{\text{total distance travelled}}{\text{time taken}} = \frac{d}{t} = \frac{\text{initial speed} + \text{final speed}}{2}$$

Example: A car travels 200 metres in 15 seconds. Calculate its average speed.

Solution:

Data:

Distance = 200 m

Time = 15

Average speed = ? $m\ s^{-1}$

Calculation:

$$\text{Average speed} = \frac{\text{distance travelled}}{\text{time taken}}$$

$$= \frac{200 \text{ metres}}{15 \text{ seconds}}$$

$$= 13.33\ m\ s^{-1}$$

The average speed of the car is 13.33 $m\ s^{-1}$.

Of course, in real life, a car would not travel at the same speed all the time. Its speed would vary according to road, traffic, weather and other conditions. To talk sensibly about things moving we need different kinds of speeds to cover different situations. These are as follows.



Constant speed is the speed of an object which is travelling the same distance in every period of time.

Average speed is the constant speed at which an object would need to travel so as to travel the same distance in the same time.

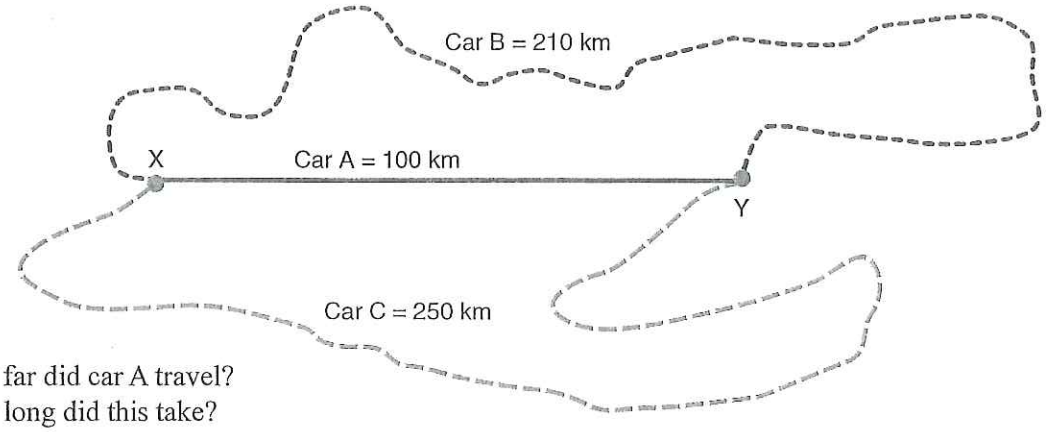
Instantaneous speed is the speed of an object in the instant of time we consider it. This will vary from instant to instant depending on, e.g. road and traffic conditions.

Initial speed is the speed of an object when we first consider it, i.e. the object's speed at the start of its journey.

Final speed is the speed of an object at the end of its journey or when we finish our consideration of its motion.

QUESTIONS

1. Consider three cars which started at town X and travelled to town Y by three different roads as shown in the diagram. Car A travelled from X to Y in 2 hours. Car B made its trip in 3 hours, while car C took 5 hours to go from X to Y.



   (a) How far did car A travel?
   (b) How long did this take?
   (c) On average, how far did car A travel each hour?
   (d) Calculate the average speed of car A.
   (e) Calculate the average speed of car B.
   (f) Calculate the average speed of car C.
   (g) Explain why we are only talking about *average* speeds here.

2. A swimmer, travelling at a steady rate, swims the 50 metre pool in 30 seconds. Calculate her average speed.
3. A rocket travels 10 000 m in 6.5 seconds. Calculate its average speed.
4. Convert to m s^{-1}:
   (a) 40 km h^{-1}
   (b) 250 cm s^{-1}
   (c) 60 km h^{-1}
   (d) 100 km h^{-1}
5. Convert to km h^{-1}:
   (a) 20 m s^{-1}
   (b) 60 m s^{-1}
   (c) 1000 cm s^{-1}
6. A racing car is attempting to break the 'standing kilometre' time record. When the starting light turns green, it accelerates at maximum rate and crosses the finish line at 180 km h^{-1} 40 s later (on a racing track of course).
   (a) Identify the initial speed of the car.
   (b) Identify its final speed.
   (c) Calculate its average speed.
   (d) Estimate its instantaneous speed 20 s after starting. Justify your answer.
   (e) Predict the constant speed to cover the same distance in the same time.

7. Fill in the missing quantities in the table.

	Distance travelled	Time taken	Average speed
(a)	1500 m	30 s	
(b)	270 m	9 s	
(c)	243 m	27 s	
(d)	12.3 m	3.2 s	
(e)	640 m	16 s	
(f)	800 m		25 m s^{-1}
(g)	300 m		12 m s^{-1}
(h)	250 km		12.5 km h^{-1}
(i)	3.6 km		12 m s^{-1}
(j)	160 km		8 km h^{-1}
(k)		3.5 hr	16 km h^{-1}
(l)		150 s	5 m s^{-1}
(m)		2 min	10 m s^{-1}
(n)		25 s	0.5 m s^{-1}
(o)		0.3 s	90 m s^{-1}

7 Displacement-Time Graphs 1

Distance-time and displacement-time graphs can be used to summarise the way an object moves. From them we can do the following.

- Read directly from the graph to find the distance travelled or the displacement of an object at particular times, or vice versa.
- Calculate the **speed** or **velocity** of the object from the **gradient** of the graph.

For example, consider the following graph.

The graph tells us (amongst other things) that:

- The object travelled 16 m in 4 seconds.
- From the gradient of the graph we get:

$$\text{Gradient} = \frac{\text{rise}}{\text{run}} = \frac{\text{distance}}{\text{time taken}}$$

$$= \text{average speed} = \frac{16}{4} = 4 \text{ m s}^{-1}$$

- As the gradient is constant, speed is constant.
- Because no direction has been given, we cannot state the velocity of this object.

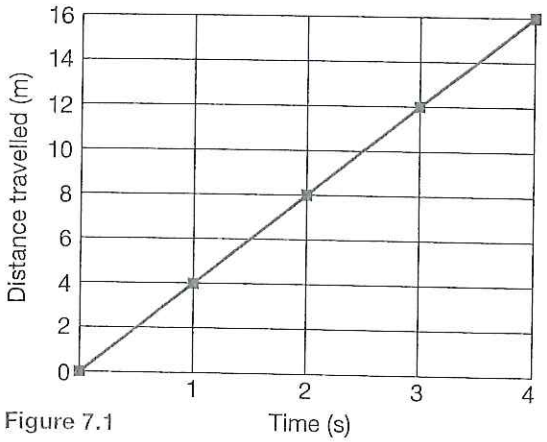


Figure 7.1

QUESTIONS

1. With reference to the graph in Figure 7.1:
   (a) How far had the object travelled after 2.5 s?
   (b) When was the object 12 m from its starting position?
   (c) What was the speed of the object at time 2.0 s?
   (d) What was the speed of the object at time 3.5 s?
   (e) Account for the similarity in your answers to (c) and (d).

Now consider the following graph.

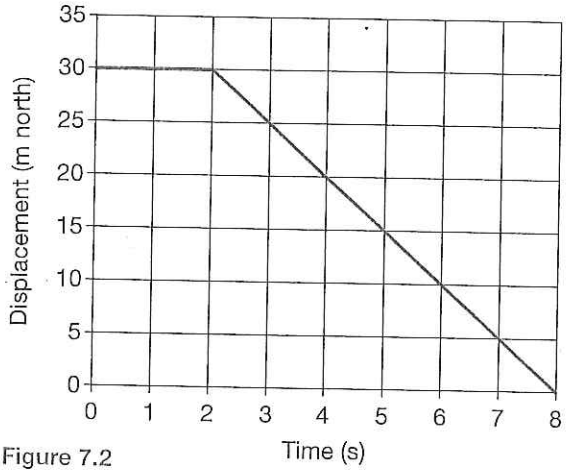


Figure 7.2

The graph tells us (amongst other things) that:

- The object was at a displacement of 30 m north at time zero and stayed there for 2 s.
- At $t = 2$ s, the object moved towards zero displacement and got there at $t = 8$ s.
- The total distance travelled by the object was 30 metres.
- The displacement of the object was 30 m south of its original position.
- It moved 30 m in 6 s.
- Its average speed while moving was therefore 5 m s^{-1} (gradient).
- Its average velocity was 5 m s^{-1} to zero displacement or 5 m s^{-1} south (gradient has a negative slope).
- Its average speed for the whole period covered was 3.75 m s^{-1} (total distance divided by *total* time), i.e.

$$\text{Average speed} = \frac{\text{total distance travelled}}{\text{total time taken}}$$

$$= \frac{30}{8} = 3.75 \text{ m s}^{-1}$$

- Its average velocity was 3.75 m s^{-1} south (displacement divided by *total* time), i.e.

$$\text{Average velocity} = \frac{\text{displacement}}{\text{time}}$$

$$= \frac{30 \text{ m S}}{8} = 3.75 \text{ m s}^{-1} \text{ south}$$

MORE QUESTIONS

2. With reference to the graph in Figure 7.2:
   (a) How far had the object travelled after 2.5 s?
   (b) What is the displacement of the object after 4 s?
   (c) When was the object 20 m from its starting position?
   (d) What was the velocity of the object at time 1.5 s?
   (e) What was the velocity of the object at time 3.5 s?
   (f) At time 7 s the object has moved 25 m south of the position it started from. Can we therefore say that its displacement at time 7.0 s is 25 m south? Explain your answer.

3. Figure 7.3 shows how the displacement of an object varied over time.

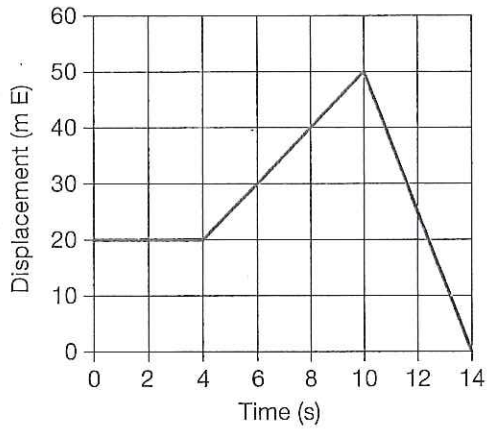


Figure 7.3

   (a) Determine the displacement of the object at time = 2, 4, 6, 8, 10, 12 and 14 s.
   (b) When was its speed constant?
   (c) Calculate the speed of the object between:
   $t = 0$ and $t = 2$ s
   $t = 4$ and $t = 10$ s
   $t = 10$ and $t = 14$ s
   (d) Calculate its velocity during these three time intervals.
   (e) Calculate the average speed of the object for the whole journey.
   (f) Calculate the average velocity for the whole trip.

4. Figure 7.4 shows the displacement-time relationship for an object over 12 seconds.

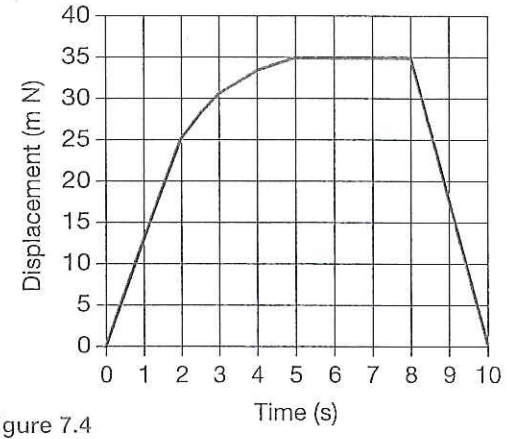


Figure 7.4

   (a) Describe the velocity of the object between time 5 and 8 seconds.
   (b) Describe the velocity of the object between time 8 and 10 seconds.
   (c) Find its average speed for the journey.
   (d) Find its average velocity for the journey.
   (e) Without drawing it, describe how a distance travelled/time graph for this object would be different to the displacement/time graph.

5. Figure 7.5 shows how the displacement of an object changes over time.

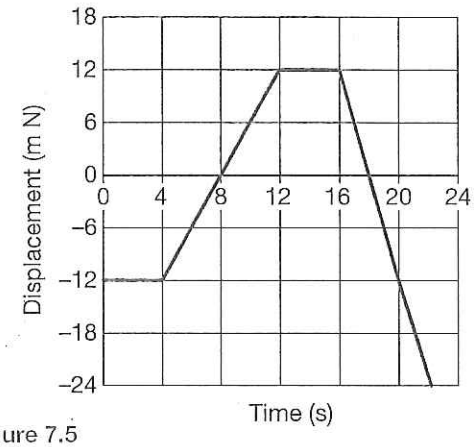


Figure 7.5

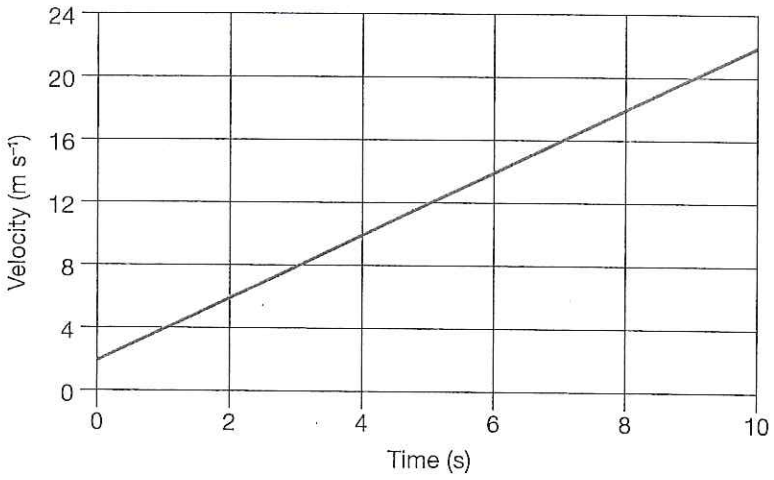
   (a) How far does the object travel in the first 8 s of its journey?
   (b) What is its average speed during the first 8 s?
   (c) What is its average velocity during the first 18 s?
   (d) Identify the instantaneous velocity of the object at $t = 9$ s.
   (e) Identify the instantaneous velocity of the object at $t = 17$ s.
   (f) What is its displacement after 22 s?
   (g) Calculate the object's average speed and velocity for the 22 s.

9 Velocity-Time Graphs 1

Velocity-time graphs can be used in the same ways as displacement-time graphs to summarise the motion of an object. From them, we can do the following.

- Read velocities or times directly from the graph.
- Use the **gradient** of the graph to find the **acceleration** of the object.
- Use the **area** under the graph to find the **change in displacement** of the object or the total distance travelled.

Example 1: The graph shows the velocity of an object as a function of time. The acceleration of the object is constant.



Among other things, this graph tells us that:

(a) The object started at a velocity of 2.0 m s^{-1} north at time zero.

(b) Its velocity increased uniformly until it was 22 m s^{-1} N at time 10 s.

(c) From $v_{av} = \frac{u+v}{2}$ the average velocity of the object was 12 m s^{-1} N.

(d) If it travelled at an average 12 m s^{-1} for 10 s, the distance travelled was 120 m.

(e) Its displacement from its starting position is therefore 120 m N.

(f) The area under the curve (straight line in this case) is 120.

(g) If we multiply the units on the y-axis by those on the x-axis we get: m s^{-1} × s = m = distance.

(h) The area under the curve of a velocity/time graph is therefore equal to the total distance travelled, and if that travel was in the one direction, it also equals the change in displacement of the object during the time given.

(i) If we now consider the gradient of the graph, then: Gradient $= \frac{\text{rise}}{\text{run}} = \frac{20 \text{ m s}^{-1}}{10 \text{ s}} = 2 \text{ m s}^{-1}$

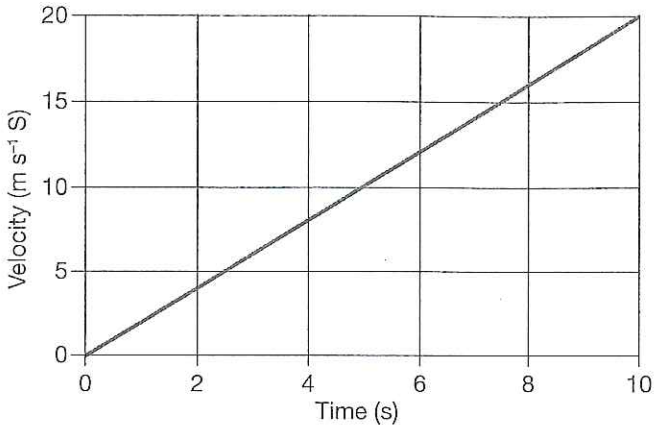
(j) Checking this using the formula: $a = \frac{v-u}{t} = \frac{22-2}{10} = 2 \text{ m s}^{-2}$

(k) So, the gradient of a velocity/time graph gives us the acceleration of the object.

Gradient = acceleration

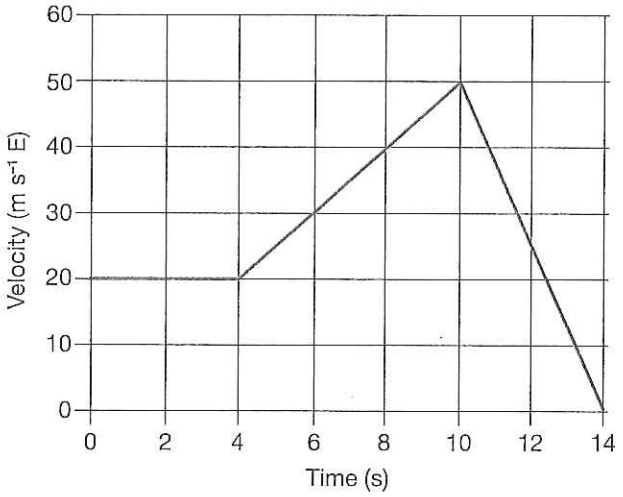
QUESTIONS

1. The graph shows how the velocity of a toy car changed over time.



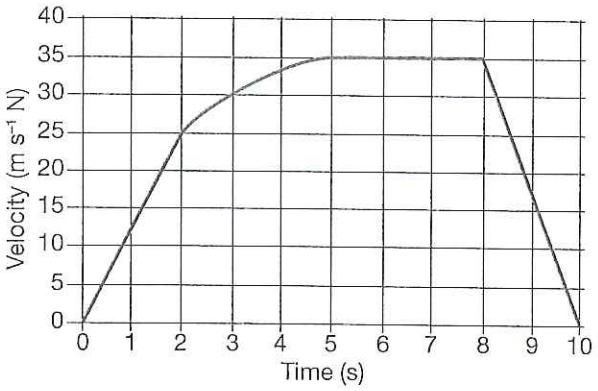
(a) Determine the initial and final velocities of the car.
(b) Calculate its acceleration.
(c) Calculate the change in displacement of the object.
(d) Calculate the total distance it travelled.
(e) Calculate its average speed.
(f) Calculate its average velocity.

2. The graph shows how the velocity of a car changed with time.



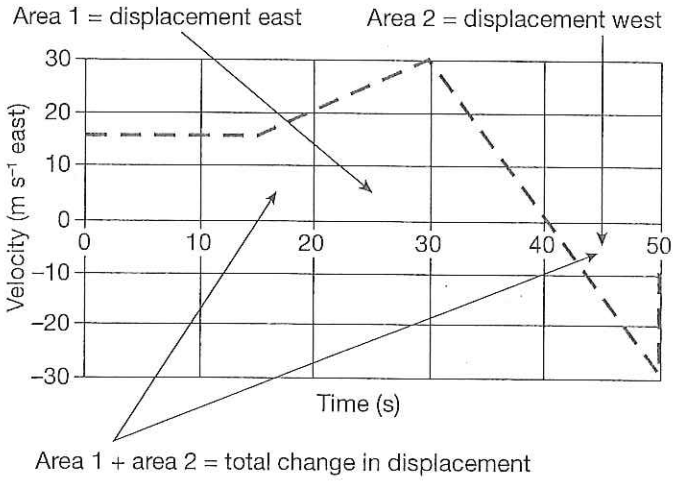
(a) Determine the velocity of the car at time $t = 6$ s.
(b) Find its acceleration at time = 6 s.
(c) Calculate the change in displacement of the car.
(d) Calculate the total distance it travelled.
(e) Calculate its average speed.
(f) Calculate its average velocity.
(g) What is its average acceleration?

3. The graph shows how the velocity of a car changed with time.



(a) Determine the velocity of the car at time $t = 3$ s.
(b) Find its acceleration at time = 9 s.
(c) Estimate the displacement of the car during the 10 s.
(d) Estimate the total distance it travelled during the 10 s.
(e) Calculate its average speed.
(f) Calculate its average velocity.
(g) What is its average acceleration?

Example 2: If the moving object changes direction, or its motion is changing with uniform acceleration, the graph is more complicated. Consider the following graph.



From this graph we can see that:

(a) The object travelled at a constant velocity of 15 m s^{-1} east for 15 seconds, then:

(b) From the gradient $= \frac{\text{rise}}{\text{run}} = \frac{\text{velocity change}}{\text{time for change}}$

$$= \frac{30 - 15}{15} = \frac{15}{15} = 1$$

It accelerated at 1.0 m s^{-2} east for the next 15 s increasing its speed to 30 m s^{-1} then:

Inertia is the property all objects have (due to their mass) that causes them to resist any attempt to change their state of uniform motion.

Inertia has significant consequences for people travelling. You feel the effect of your inertia when the school bus turns a corner, as a plane takes off or as a car brakes suddenly.

Newton's first law of motion is about inertia – 'Any object at rest or in uniform motion in a straight line will remain at rest or in uniform motion until a net force acts on it'.

QUESTIONS

1. Outline the dangers presented by loose objects in a moving vehicle and when these dangers can occur.
2. Explain why unrestrained passengers are at risk in moving vehicles.
3. Assess the danger to an unbelted driver in a moving car compared to an unbelted passenger.
4. Explain how seatbelts help prevent injury to people in vehicles that stop unexpectedly and suddenly.
5. Explain the role of the inertia of passengers in each of the following situations. For each situation, outline what forces act on the passenger (ignore weight force and reaction to weight force).
   (a) The school bus turning a corner.
   (b) A plane accelerating down the runway.
   (c) A lift in a high-rise building starting to move upwards.
   (d) A lift in a high-rise building starting to move downwards.
6. Assess the effectiveness of lap, lap-sash and harness seatbelts in reducing the effects of collisions.
7. Assess reasons for introducing lower speed limits for cars in built-up areas.
8. The speed limit on the open road is 110 km h^{-1}. On some straight roads in the country this seems unrealistically low. Assess reasons for the limit.
9. Outline how airbags work to reduce the effects of inertia.
10. Use the principles of physics to explain how crumple zones reduce the effects of inertia in collisions.
11. Imagine identical cars travelling at 50, 60, 80 and 100 km h^{-1}.
    (a) Calculate the ratio of their kinetic energies.
    (b) Find the ratio of the work brakes need to do to stop the cars.
    (c) The cars all crash into a tree and stop in 0.5 s. Find the ratio of the forces on each car.
    (d) Calculate how the magnitudes of the forces would be affected if the collision time was 0.1 s.
    (e) List features of modern cars that are designed to reduce the effects of collisions like this.
12. A car crashes and stops from 60 km h^{-1} in 0.1 s. Calculate:
    (a) The deceleration of the car.
    (b) The force a mother would need to apply to a 10 kg child she was nursing (unrestrained) in order to overcome inertial effects.
    (c) The equivalent mass she would be able to lift if she could do this.
    (d) Assess the feasibility of this.
13. The inertia of a spacecraft in deep space keeps it travelling towards its destination. The inertia of a moving car does not – it slows and stops. Explain the reasons for the difference.
14. When you are in a car that turns a corner quickly, you feel as if you are 'thrown to the side'. Explain what is really happening and why 'thrown' is a totally incorrect way to describe this observation.

22 Newton's Force Equation

The acceleration of an object depends on its mass and on the size of the force applied to it. This can be expressed in the form of an equation.

$F = ma$	Where	F = force in newtons
		m = mass in kilograms
		a = acceleration in m s^{-2}

This relationship was first discovered by Isaac Newton, and is called **Newton's second law of motion.** It can be stated as follows.

> When an unbalanced force acts on an object it produces an acceleration that is directly proportional to the force and indirectly proportional to the mass of the object.

Notice, that from the equation, we can get a definition for the newton, the unit of force. Imagine we have a mass of 1 kg and we wish to accelerate it at 1 m s^{-2}. From the equation, it is obvious that a force of 1 N is needed.

$F = m \times a$

$1 = 1 \times 1$

1 N = 1 kg × 1 m s^{-2}

Or, put another way, 1 newton = 1 kg m s^{-2}.

> One newton is the force needed to give a mass of 1 kg an acceleration of 1 m s^{-2}.

Newton's force equation can be used, like all our other equations, to predict the outcomes of situations without having to do experiments.

Example: A toy car, mass 4 kg, is pushed by a force of 12 newtons. What acceleration results?

Solution:	Data:	Calculation:
	$m = 4$ kg	From $F = ma$
	$F = 12$ N	$12 = 4a$
	$a = ?$	So $a = 12/4 = 3$ m s^{-2}

Answer: Acceleration on the car is 3 m s^{-2}.

QUESTIONS

1. Calculate the force that will give a toy car of 2.5 kg an acceleration of 6.0 m s^{-2}.
2. Calculate the force that gives a toy car of 4.0 kg an acceleration of 1.25 m s^{-2}.
3. A trolley of mass 2.5 kg is accelerating at 12 m s^{-2}. Calculate the force on it.
4. Calculate the acceleration of a 2.5 kg trolley if a force of 37.5 N acts on it.
5. A force of 13 N acts on a 4.2 kg trolley. Calculate the acceleration produced.
6. A force of 36 N acts on a 3 kg trolley. Calculate the acceleration produced.
7. Calculate the mass that would accelerate at 2 m s^{-2} if a force of 500 N acted on it.
8. 64 N gives a car an acceleration of 0.5 m s^{-2}. Calculate its mass.
9. A 3.5 kg toy car is accelerating at 4.0 m s^{-2}. Calculate the force acting on it.
10. A 250 g car has a force of 0.75 N acting on it. Calculate its acceleration.
11. Use the formula to find the missing quantities in the table.

Force applied (N)	Mass (kg)	Acceleration (m s^{-2})
12	3	A
24	6	B
150	7.5	C
45	D	5
27	E	3
18	F	4.5
G	7	3
H	1.5	5
I	0.5	14

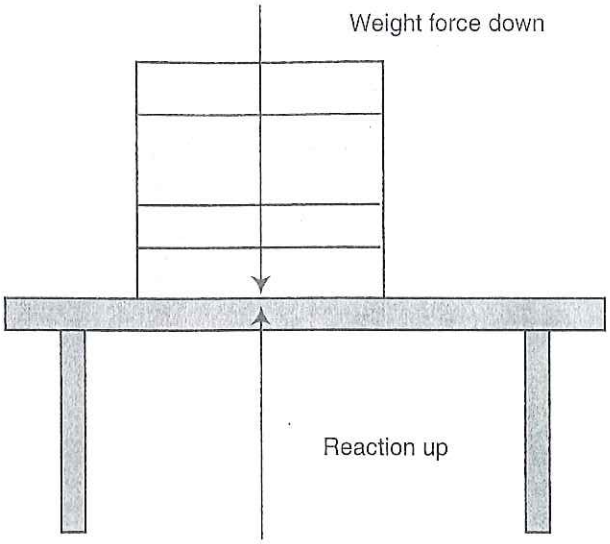
12. A force of 1.5 N acts on a 300 gram body. Calculate its acceleration.
13. Find the mass that would be given an acceleration of 0.2 m s^{-2} if a force of 42 N acted on it.
14. A force of 0.5 N accelerates an object at 0.25 m s^{-2}. Calculate its mass.
15. A 7.0 kg object is acted on by a force that gives it an acceleration of 0.5 m s^{-2}. Calculate the force causing this acceleration.

24 Newton's Third Law

When one object puts a force on another, the second always exerts an equal and opposite force back on the first. We call these forces an **action force** and a **reaction force**.

Action and reaction forces always act on **different objects**. If they were to act on the same object we would never have acceleration because the net force on objects would always be zero.

A simple example of an action-reaction pair is a pile of books on a table as shown in the diagram. The weight force of the books is supported by the upwards reaction force of the table. Gravity pulls them down and the table holds them up.



Newton's third law states that:

For every action, there is an equal and opposite reaction.

An alternate expression of Newton's third law comes from our study of momentum changes (see later work). This considers the forces objects apply to each other during collisions and the times for which these forces act. For momentum to be conserved (as it must be in the absence of external forces), the change in momentum of one object involved in the collision (equal to the **impulse** applied to it) must be equal and opposite to the change in momentum of the other object, or as follows.

For every impulse, there is an equal and opposite impulse.

QUESTIONS

1. Briefly describe how each of the following illustrates Newton's third law, and draw a rough diagram showing the forces involved in each.
   (a) A weightlifter holding weights above his head.
   (b) A tennis ball being hit.
   (c) A person walking.
   (d) A boat being rowed across a lake.
   (e) A boxer punching a punching bag.
2. An astronaut was fixing the Hubble telescope when she lost her grip. Her safety line came undone and she floated 50 m out into space. All she had is her tool belt with its tools around her waist and oxygen for 2 hours. Explain how she can use Newton's third law to return to the spaceship.
3. A block of wood is pushing down on a desk with a force of 15 N. Identify any other force(s) present in this situation.
4. Identify the action-reaction pairs in each of the following situations.
   (a) Cricket ball hitting cricket bat.
   (b) Rocket taking off.
   (c) Soccer player kicking soccer ball.
   (d) Hammer hitting nail.
   (e) Ball bouncing on ground.
   (f) A horse pulling a cart along.
   (g) Person punching a punching bag.
5. If a horse pulls a cart, then the horse applies a force on the cart. Explain how this system can move if the cart puts an equal and opposite force on the horse. Don't these two forces simply cancel each other?